📚 STUDENT INSTRUCTION SHEET

SQL + Python Project using CRISP-DM Framework

🎯 Objective of the Project

You are acting as a **Business Analyst + Data Engineer**.
Your task is to convert a **real-world business problem (given case)** into:

- A structured **database system (SQL)**
- Analytical insights using **Python**

You must follow the **CRISP-DM methodology** throughout the project.

Each case naturally maps to **CRISP-DM stages**:

- **Business Understanding:** Problem is implicit, not explicit
- **Data Understanding:** Students identify entities & attributes
- **Data Preparation:** Schema design (tables, keys)
- **Modeling:** SQL queries + Python analytics
- **Evaluation:** Insights & validation
- **Deployment:** Reports/dashboard

CRISP-DM STAGES (What You Must Do)

Stage 1: Business Understanding

🎯 Goal:

Understand the business problem (DO NOT jump to tables)

✓ Your Tasks:

- Read the case carefully

- Identify:
  - What is the organization trying to achieve?
  - What problems are hinted at?
  - What decisions need support?

📘 Deliverable:

- 2–3 page document including:
  - Problem statement (in your own words)
  - At least **5 business questions**
  - Assumptions made
  - Expected outcomes

💭 Example Thinking:

"Why are some products not selling?"
"Which customers contribute most revenue?"

● Stage 2: Data Understanding

🎯 Goal:

Identify what data is needed to answer business questions

✓ Your Tasks:

- Identify **entities (objects)** in the system
- List attributes for each entity
- Identify relationships between entities

📘 Deliverable:

- Entity list with attributes
- Relationship explanation
- ER Diagram

⚠ Important:

Do NOT think in SQL yet. Think in **real-world objects**.

Stage 3: Data Preparation

🎯 Goal:

Convert conceptual model into database-ready structure

✔ Your Tasks:

- Convert ER Diagram → Tables
- Define:
  - Primary Keys
  - Foreign Keys
  - Data Types
  - Constraints
- Insert **50–100 records**

📘 Deliverable:

- SQL schema (CREATE statements)
- Data insertion scripts

💡 Tip:

Ensure **data consistency and realism**

Stage 4: Modeling (SQL Queries)

🎯 Goal:

Extract meaningful information using SQL

✔ Your Tasks:

Basic Queries:

- SELECT, WHERE, ORDER BY

Intermediate:

- JOIN (minimum 3)
- GROUP BY, HAVING

Analytical:

- Top performers (customers/products/etc.)
- Trends (time-based)
- At least 2 subqueries

📘 Deliverable:

- SQL query file with comments explaining each query

💡 Focus:

Every query should answer a **business question**

🔴 Stage 5: Evaluation (Insights & Reports)

🎯 Goal:

Translate query output into business insights

✔ Your Tasks:

- Create at least **3 reports**
- Interpret results:
  - What does the data say?
  - Why does it matter?

📄 Deliverable:

- Report document with:
  - Output (tables/graphs)
  - Explanation of insights

💡 Example:

"Sales peak during weekends → staffing needs adjustment"

Stage 6: Deployment (Python Integration)

🎯 Goal:

Use Python to perform analysis beyond SQL

✔ Your Tasks:

- Connect Python to your database
- Extract data using SQL
- Perform analysis using:
  - pandas
  - matplotlib

📄 Deliverable:

- Python notebook/script including:
  - Data extraction
  - At least 1 visualization
  - Explanation of results

💡 **Example:**

- Trend analysis
- Customer segmentation
- Frequency patterns

FINAL SUBMISSION CHECKLIST

Before submitting, ensure you have:

✔ Business Understanding Document
✔ ER Diagram
✔ SQL Schema (CREATE statements)
✔ Data Inserts (50–100 records)
✔ SQL Queries (with explanations)
✔ Reports with insights
✔ Python Notebook
✔ Presentation (8–10 mins)

Each group must submit:

📁 Folder Structure

```
Project_Name/
│
├── 1_Business_Understanding.pdf
├── 2_ER_Diagram.png / pdf
├── 3_SQL_Schema.sql
├── 4_Data_Inserts.sql
├── 5_SQL_Queries.sql
├── 6_Reports.pdf
├── 7_Python_Code.ipynb
└── 8_Presentation.pptx
```

⚠ IMPORTANT GUIDELINES

- ✖ Do NOT copy schemas from the internet
- ✓ Every table must be justified

- ✓ Focus on solving a business problem
- ✓ Keep design simple (max 6–7 tables)
- ✓ Think like a consultant, not just a coder

HOW YOU WILL BE EVALUATED

You are NOT evaluated on:

- Fancy diagrams
- Complex code

You ARE evaluated on:

- Logical thinking
- Business understanding
- Clarity of insights

💬 FINAL THOUGHT

👉 *"Data is useful only when it supports a decision."*

Your project should answer:
"What action can the business take based on this system?"

📊 PROJECT EVALUATION RUBRIC (100 MARKS)

Component	Marks	Excellent (A)	Good (B)	Average (C)	Poor (D)
Business Understanding	10	Clear, deep interpretation of problem; strong business questions; insightful assumptions	Good understanding with relevant questions	Basic understanding; limited interpretation	Weak, unclear or copied
ER Diagram & Data Understanding	10	Accurate entities, relationships, well-structured ER diagram	Minor errors in relationships or attributes	Some missing/incorrect relationships	Poor or incomplete diagram
Database	15	Well-	Mostly correct	Errors in	Weak or

Design (Schema)		normalized tables; correct PK, FK, constraints	structure; minor issues	keys/relationships	incorrect schema
Data Preparation	10	Realistic, diverse, consistent dataset (50–100+ records)	Mostly realistic with minor issues	Limited or repetitive data	Poor or insufficient data
SQL Queries (Technical + Analytical)	20	Complex queries; joins, subqueries, aggregation; business insights derived	Good variety of queries	Basic queries only; limited analysis	Incorrect or minimal queries
Reports & Insights	10	Insightful, clearly linked to business decisions	Some useful insights	Descriptive outputs without insights	No meaningful interpretation
Python Integration	15	Strong integration; data extraction + meaningful analysis + visualization	Functional analysis; basic visuals	Minimal use of Python	Poor or missing
Presentation	10	Clear, structured, confident; strong storytelling	Good presentation with minor gaps	Basic explanation	Unclear or poorly delivered